

In the Name of God, the Creator of All Crafts

Management Information Systems (MIS) Project

Phase 2: Database Design & Implementation in SQL Server

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1 Phase 2 Objective: Enterprise Data Architecture

The primary objective of Phase 2 is to transform the process models engineered in previous phases into a robust, normalized relational database. This involves refining, extending, and synthesizing baseline factory operational data within Excel, implementing the schema inside **Microsoft SQL Server**, and executing analytical SQL queries that directly drive the management reporting dashboards planned for Phase 3.

2 Dataset Enrichment & Synthetic Generation

Students are provided with a baseline semi-structured workbook, `MIS.xlsx`, containing incomplete operational datasets (e.g., shift logs, processing plant units, material flow streams). You are required to audit, enrich, and adapt these data tables to reflect the copper extraction workflows modeled in Phase 1.

2.1 Mandatory Rules for Dataset Preparation

1. Process-to-Data Alignment:

- Review all baseline tables and align them strictly with your modeled workflows. Records and tables must be modified, purged, or expanded accordingly.
- Students may refer to the supplementary industrial reference dataset provided on the course learning portal to acquire deeper domain context regarding equipment rates, assay grades, and material balances.

2. Schema Expansion (Tables & Custom Attributes):

- The finalized relational dataset must contain **at least ten (10) distinct tables**.
- You must engineer **at least six (6) entirely new, custom attributes (columns)** that do not exist in the initial template, formulated based on your process analysis.

3. Synthetic Data Generation & Operational Realism:

- Generate **at least 150 rows of data per table**.
- Data values must be operationally realistic, adhering to industrial chemical, physical, and operational distributions (e.g., equipment recovery rates, head grades, conveyor tonnage).
- **Temporal Bound:** All generated timestamps and operational dates must fall strictly within a **maximum time window of two (2) years**.
- Data distributions should foster statistical correlation, enabling meaningful pattern extraction and multivariate analysis across operations.

3 Relational Modeling & Database Implementation

Bonus Component: Entity-Relationship Diagram (ERD) Design

Before physical implementation, you are encouraged to formulate the conceptual and logical schema:

- Define all operational **Entities** and their corresponding **Attributes**.
- Specify structural **Relationships**, Primary Keys (PK), Foreign Keys (FK), and referential integrity rules.
- Explicitly map cardinalities ($1 : 1$, $1 : N$, $M : N$).
- *Tooling*: Any specialized modeling software may be utilized (**Microsoft Visio** is recommended).

Note: While marked as optional/bonus, a comprehensive ERD significantly streamlines physical database construction and query optimization.

3.1 Physical Implementation in Microsoft SQL Server

Construct the relational schema within a dedicated Microsoft SQL Server database:

- Create normalized tables with appropriate SQL data types (INT, VARCHAR, DECIMAL, DATETIME, etc.).
- Enforce integrity constraints: PRIMARY KEY, FOREIGN KEY, NOT NULL, and CHECK constraints.
- Establish foreign key relationships matching the process interactions modeled in Phase 1.

3.2 Data Migration (Excel to SQL Server)

- Migrate the finalized tabular sheets from the Excel workbook into their designated SQL Server destination tables (e.g., using SQL Server Import and Export Wizard or bulk insert scripts).
- Ensure strict schema compatibility between Excel column types and target SQL column definitions to prevent truncation or conversion faults.

4 Analytical SQL Query Formulation

You must author and execute **at least eight (8) meaningful, non-trivial SQL queries** designed to extract business and operational intelligence from the copper extraction lifecycle. These queries must directly establish the data layer for the Phase 3 executive dashboards.

Illustrative Analytical Query Domains

- **Metallurgical Recovery Assays**: Computing weighted average copper grade/purity across distinct chemical flotation and electrorefining batches.
- **Process Efficiency & Recirculation**: Identifying comminution or screening units exhibiting greater than 10% recirculating scrap/tailings load.
- **Operational Bottlenecks**: Calculating mean equipment downtime per shift categorized by failure classification.

5 Supplementary Report Specifications

The project submission must include a concise, insight-driven technical report covering:

1. **Data Dictionary:** Comprehensive definition of all final tables and attributes, with particular focus on custom attributes introduced by the student.
2. **Query Rationale:** Technical description and operational objective of each of the 8+ selected analytical SQL queries.
3. **Engineering Assumptions:** Explicit documentation of all working assumptions, including relationship cardinalities, domain constraints, data type sizing, and simulated operational efficiencies.

Note: Voluminous, verbose documentation is discouraged. Focus on technical rigor, clarity of insights, and systemic understanding.

6 Submission Package Format

Deliverables must be bundled into a single compressed **ZIP** archive containing:

1. Finalized, populated Excel workbook (.xlsx).
2. High-resolution screenshots of the relational tables created in SQL Server Management Studio (SSMS).
3. Final ERD schematic (as an export image or native diagram source file).
4. High-resolution, full-screen captures of all SQL query scripts alongside their corresponding output result grids.
5. Supplementary Technical Report provided in both **Word** (.docx) and **PDF** (.pdf) formats.

Crucial: All submitted graphical captures must be full-screen, clean, and legibly formatted.